



PREDICT DATA ANALYTICS SYSTEM

(A Machine Learning & Data Analytics Project)

Internship Project Report

Submitted By

Akshay Kumar

B.Tech in Data Science

Intern at

Readynest Corp

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Readynest Corp

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CERTIFICATE

This is to certify that Mr. Akshay Kumar, a student of B.Tech in Data Science, has successfully completed the internship project titled “Predict Data Analytics Systems” during his internship at ReadyNest Corp.

This Project was carried out under proper guidance and demonstrates the practical implementation of Data Analytics, Machine Learning, and Dashboard Development techniques. The work presented in this report is original and has been completed during the internship period.

We appreciate his dedication, hard work, and technical skills demonstrated throughout the internship and wish him success in all his future endeavors.

Project Mentor

ReadyNest Corp

Authorized Signatory

ReadyNest Corp

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to ReadyNest Corp for providing me with the opportunity to work on the Predict Data Analytics System during my internship.

I am especially thankful to my project mentor and the entire ReadyNest Corp team for their valuable guidance, continuous support, and encouragement throughout the internship. Their mentorship helped me gain practical knowledge in Data Analytics, Machine Learning, and Dashboard Development.

I also express my heartfelt thanks to the faculty members of CGC University for their continuous motivation and support throughout my academic journey.

Finally, I would like to thank my parents, friends, everyone who directly or indirectly contributed to the successful completion of this internship project.

ABSTRACT

The Predict Data Analytics System is a Machine Learning and Data Analytics project developed during my internship at ReadyNest Corp. The primary objective of this project is to analyze Flipkart product reviews and predict customer ratings based on review text using Machine Learning techniques.

The project starts with data collection and preprocessing, where duplicate records, missing values, and unnecessary information are removed to prepare a clean dataset. Exploratory Data Analysis (EDA) is then performed to understand customer review patterns, rating distributions, and product performance.

A Logistic Regression model combined with TF-IDF Vectorization is used to predict product ratings from textual reviews. The trained model achieved an accuracy of approximately 73%, demonstrating its effectiveness in review classification.

An interactive dashboard was developed using Streamlit to provide dataset exploration, data visualization, rating prediction, and business insights. The dashboard enables users to interact with the dataset and understand customer feedback in a simple and intuitive manner.

This project demonstrates how Data Analytics and Machine Learning can help organizations analyze customer opinions, improve products, and support data-driven business decisions.

Keywords:

Data Analytics, Machine Learning, Logistic Regression, TF-IDF, Streamlit, Product Reviews, Rating Prediction.

TABLE OF CONTENTS

1. Introduction
2. Problem Statement
3. Objectives
4. Dataset Description
5. Methodology
6. Data Collection
7. Data Cleaning
8. Exploratory Data Analysis (EDA)
9. Data Visualization
10. Model Training
11. Rating Prediction
12. Business Insights
13. Dashboard Development
14. Results and Discussion
15. Conclusion
16. Future Scope
17. References

CHAPTER 1

INTRODUCTION

In today's digital world, e-commerce platforms generate a huge amount of customer reviews every day. These reviews help customers make purchasing decisions and provide valuable feedback to businesses about their products and services. However, manually analyzing thousands of customer reviews is a difficult and time-consuming task.

Data Analytics and Machine Learning provide effective solutions for extracting meaningful information from customer reviews. By analyzing review text, businesses can identify customer satisfaction, common issues, and product quality. Machine Learning models can automatically predict customer ratings based on review content, reducing manual effort and improving decision-making.

The Predict Data Analytics System is designed to analyze Flipkart product reviews using Data Analytics techniques and Machine Learning algorithms. The system performs data preprocessing, exploratory data analysis, visualization, and rating prediction using a Logistic Regression model with TF-IDF Vectorization.

An interactive dashboard developed using Streamlit allows users to explore the dataset, visualize review trends, predict ratings, and gain business insights. This project demonstrates the practical application of Data Analytics and Machine Learning in solving real-world business problems.

CHAPTER 2

PROBLEM STATEMENT

In today's digital era, e-commerce platforms receive thousands of customer reviews every day. These reviews contain valuable information about customer experiences, product quality, and satisfaction. However, manually analyzing such a large volume of reviews is time-consuming, inefficient, and prone to human error.

Businesses often struggle to identify customer sentiment, recognize common product issues, and understand overall customer satisfaction from unstructured textual data. Without proper analysis, it becomes difficult to make informed business decisions and improve product quality.

The primary challenge is to automatically analyze customer reviews and predict product ratings based on review text. An efficient system is required that can clean the dataset, visualize customer feedback, generate meaningful insights, and accurately predict ratings using Machine Learning techniques.

The Predict Data Analytics System addresses this problem by using Data Analytics and Machine Learning to automate review analysis, predict ratings, and provide business insights through an interactive Streamlit dashboard.

CHAPTER 3

OBJECTIVES

Objectives of the project

The main objectives of the Predict Data Analytics system are:

- To collect and analyze FlipKart product review data.
- To clean and preprocess the dataset for accurate analysis.
- To perform Exploratory Data Analysis (EDA) to identify review patterns.
- To visualize customer feedback using interactive charts.
- To develop a Machine Learning model for product rating prediction.
- To predict customer ratings based on textual reviews.
- To generate business insights from customer reviews.
- To build an interactive Streamlit dashboard for easy data exploration and prediction.

CHAPTER 4

DATASET DESCRIPTION

4.1 Dataset Source

The dataset used in this project was collected from Flipkart Product Reviews. It contains customer reviews, product names, and product ratings. The dataset was selected because it provides real-world customer feedback, which is suitable for applying Data Analytics and Machine Learning techniques.

4.2 Dataset Overview

The dataset contains customer reviews of various products available on the Flipkart platform. Each record represents one customer review along with the product name and the rating given by the customer.

The original dataset contained **2304 records**. After removing duplicate records and unnecessary columns during the data cleaning process, the final dataset consisted of **2181 records**

4.3 Dataset Features

Column Name	Description
Product_name	Name of the product
Review	Customer review text
Rating	Customer rating (1–5 Stars)

4.4 Dataset Statistics

Attribute	Value
Original Records	2304
Cleaned Records	2181
Number of Features	3
Target Variable	Rating
Rating Range	1 – 5

4.5 Target Variable

The target variable used for prediction is Rating.

The Machine Learning model predicts the customer rating based on the review text.

Ratings range from 1 (Very Poor) to 5 (Excellent)

